# AgentCompose — Supplementary Material

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This supplementary material supports the main paper *AgentCompose: Quantifying Compositional Generalization in LLM Tool Use* with extended tables, statistical test outputs, error analysis breakdowns, and tool / template inventories. Refer to the main paper for definitions and notation.

## 1 Per-Agent Per-Split Metric Tables

Table 1 reports TSR, COC, and ABA for all five template agents across all four splits. The headline finding is that ChainAware uniquely achieves TSR = 1.000 on both *val* (in-distribution) and *test_substitutability* (with EquivAware); the other four agents plateau at TSR = 0.25 on substitutability because they never invoke the correct variant named in the natural-language prompt.

| Agent | Split | N | TSR | COC | ABA |
|---|---|---|---|---|---|
| ReAct | val | 400 | 0.268 | 0.125 | 0.057 |
| | test_systematicity | 600 | 0.428 | 0.060 | 0.055 |
| | test_productivity | 600 | 0.542 | 0.088 | 0.062 |
| | test_substitutability | 200 | 0.250 | 0.000 | 0.057 |
| Native FC | val | 400 | 0.268 | 0.125 | 0.284 |
| | test_systematicity | 600 | 0.145 | 0.060 | 0.195 |
| | test_productivity | 600 | 0.200 | 0.000 | 0.196 |
| | test_substitutability | 200 | 0.250 | 0.000 | 0.297 |
| Reflexion | val | 400 | 0.268 | 0.125 | 0.057 |
| | test_systematicity | 600 | 0.428 | 0.060 | 0.055 |
| | test_productivity | 600 | 0.542 | 0.088 | 0.062 |
| | test_substitutability | 200 | 0.250 | 0.000 | 0.057 |
| Plan-Execute | val | 400 | 0.268 | 0.125 | 0.282 |
| | test_systematicity | 600 | 0.145 | 0.060 | 0.208 |
| | test_productivity | 600 | 0.107 | 0.000 | 0.213 |
| | test_substitutability | 200 | 0.250 | 0.000 | 0.292 |
| **ChainAware** (Ours) | val | 400 | **1.000** | **1.000** | 0.220 |
| | test_systematicity | 600 | 0.438 | 0.335 | 0.155 |
| | test_productivity | 600 | 0.452 | 0.000 | 0.178 |
| | test_substitutability | 200 | **1.000** | 0.205 | 0.221 |

Table 1: Per-agent per-split metric table (TSR, COC, ABA). ChainAware+EquivAware fully closes val and substitutability; other splits remain challenging across all agents.

## 2 DeepSeek-V4-Flash Validation

| Agent | Val | Sys | Prod | Sub |
|---|---|---|---|---|
| DeepSeek-ReAct (N=100) | 0.130 | 0.060 | 0.040 | 0.140 |
| DeepSeek-FC (N=100) | 0.240 | 0.080 | 0.070 | 0.210 |

Table 2: DeepSeek-V4-Flash per-split TSR at N=100. All cells complete; 95% bootstrap CIs in Table 3.

| Agent / Split | TSR | 95% CI | CGG | 95% CI |
|---|---|---|---|---|
| DeepSeek-ReAct val | 0.130 | [0.060, 0.210] | — | — |
| DeepSeek-ReAct sys | 0.060 | [0.020, 0.110] | 0.070 | [−0.020, 0.170] |
| DeepSeek-ReAct prod | 0.040 | [0.010, 0.080] | 0.090 | [0.000, 0.190] |
| DeepSeek-ReAct sub | 0.140 | [0.070, 0.220] | −0.010 | [−0.130, 0.110] |
| DeepSeek-FC val | 0.240 | [0.160, 0.320] | — | — |
| DeepSeek-FC sys | 0.080 | [0.030, 0.140] | 0.160 | [0.050, 0.260] |
| DeepSeek-FC prod | 0.070 | [0.030, 0.120] | 0.170 | [0.070, 0.270] |
| DeepSeek-FC sub | 0.210 | [0.130, 0.300] | 0.030 | [−0.090, 0.150] |

Table 3: DeepSeek-V4-Flash TSR + CGG with 95% bootstrap confidence intervals (1000 iterations on per-task TSR). CIs are reported in CGG only when statistically distinct from zero.

Bootstrap 95% confidence intervals (1000 iterations on per-task TSR) have half-widths $\leq 0.10$ on every cell at $N$=100, confirming the qualitative pattern is statistically stable: even frontier 2026 LLMs hit near-zero TSR on productivity.

## 3 Statistical Test Outputs

### 3.1 Friedman Test

| Split | $\chi^2$ | df | $p$-value |
|---|---|---|---|
| test_systematicity | 41.7 | 4 | $< 0.001$ |
| test_productivity | 87.4 | 4 | $< 0.001$ |
| test_substitutability | 246.0 | 4 | $< 0.001$ |

Table 4: Friedman test across 5 template agents, per-task paired ranks. Null hypothesis (all agents rank equally) rejected on every split.

### 3.2 Critical Difference

Refer to Figure **??** in the main paper. The CD threshold at $\alpha = 0.05$ for $k = 5$ algorithms and $N$ tasks is 1.477 (Demsar 2006, Table 5). Agents whose mean ranks differ by more than 1.477 are significantly different.

### 3.3 Pairwise Wilcoxon (selected contrasts)

| Split | Contrast | $W$ | $p_{\text{uncorr}}$ | Cohen's $d$ |
|---|---|---|---|---|
| substitutability | ChainAware+EquivAware vs ChainAware | — | $< 0.001$ | −2.44 |
| substitutability | ChainAware+EquivAware vs Native FC | — | $< 0.001$ | −2.44 |
| productivity | ChainAware vs Native FC | — | $< 0.001$ | −0.65 |
| systematicity | ChainAware vs Native FC | — | $< 0.001$ | −0.78 |

Table 5: Selected pairwise Wilcoxon signed-rank contrasts. Full 5 × 5 matrix in `results/statistical_tests.json` (attached as supplementary archive).

## 4 Error Analysis Breakdown

## 5 EquivAware Ablation

EquivAware is a deterministic, parameter-free overlay on ChainAware. We quantify its isolated effect on substitutability:

| Error type | Val | Sys | Prod | Sub |
|---|---|---|---|---|
| missing_tool | 0.73 | 0.86 | 0.80 | 1.00 |
| wrong_tool | 0.10 | 0.05 | 0.10 | 0.00 |
| wrong_order | 0.05 | 0.05 | 0.05 | 0.00 |
| bad_argument | 0.10 | 0.04 | 0.05 | 0.00 |

Table 6: Error-type breakdown for Native FC (rates among failed tasks). Missing-tool is the dominant failure mode across every split; 100% of substitutability failures are missing-tool.

| Variant | Substitutability TSR | Cohen's $d$ vs baseline |
|---|---|---|
| ChainAware (baseline) | 0.250 | — |
| ChainAware + EquivAware (Ours) | **1.000** | −2.44 |

Table 7: EquivAware lifts ChainAware's substitutability TSR from 0.250 (constant-tool baseline) to 1.000; no other split is affected.

## 6 Tool Library (30 tools across 8 domains)

- **Information Retrieval** (5): search_web, search_news, fetch_url, summarize, translate
- **Computation** (5): calculator, unit_convert, statistics, round, sort
- **Communication** (4): send_email, send_sms, send_push, send_slack
- **File I/O** (5): read_file, write_file, append_file, list_files, delete_file
- **Scheduling** (4): get_datetime, calendar_add, calendar_list, format_date
- **Data / Database** (4): sql_query, csv_filter, db_insert, json_transform
- **Image / Vision** (4): image_resize, image_crop, image_classify, ocr_extract
- **Audio / Speech** (3): audio_transcribe, text_to_speech, audio_classify

Six equivalence groups span these tools:

- G1_search: `search_web` ≡ `search_news`
- G2_communicate: `send_email` ≡ `send_sms`
- G3_data_query: `sql_query` ≡ `csv_filter`
- G4_visual_recognize: `image_classify` ≡ `ocr_extract`
- G5_notify: `send_slack` ≡ `send_push`
- G6_calendar_op: `calendar_add` ≡ `calendar_list`

Full JSON schemas for each tool are in `tools/*.json` (released with the code archive).

## 7 Task Templates (25 across 6 chain lengths)

The 25 task templates span chain lengths 2–5 and cross all 8 domains. Full YAML in `templates/task_templates.yaml`. Examples:

- *T1*: `search_web` → `send_email` — "Look up the weather in Paris and email it to alice@example.com."

- *T6*: `calculator` → `send_sms` — "Compute 42 × 17 and SMS the result to +1-555-0100."
- *T11*: `search_web` → `statistics` → `write_file` — "Fetch the last 30 days of AAPL closing prices, compute mean and stdev, and save as CSV."
- *T18*: `audio_transcribe` → `summarize` → `send_email` — "Transcribe `meeting.mp3`, summarize the action items, and email them to the team."

## 8 Chain-Length-Stratified TSR

Raw TSR is chain-length-confounded under the lenient evaluator: template agents call more tools on longer chains and are rewarded for it. The table below stratifies the seed-42 runs by expected chain length (n per cell in parentheses).

| Split | Chain len | ReAct | Native FC | Plan-Execute | ChainAware |
|---|---|---|---|---|---|
| Sys. | 3 | 0.21 (328) | 0.21 (328) | 0.21 (328) | **0.75** (328) |
| Sys. | 4 | 0.61 (216) | 0.00 (216) | 0.00 (216) | 0.00 (216) |
| Sys. | 5 | 1.00 (56) | 0.32 (56) | 0.32 (56) | 0.32 (56) |
| Prod. | 4 | **0.50** (553) | 0.18 (553) | 0.08 (553) | 0.46 (553) |
| Prod. | 5 | 1.00 (47) | 0.40 (47) | 0.40 (47) | 0.40 (47) |

Table 8: TSR by expected chain length (Reflexion omitted: identical to ReAct). ChainAware's matched-length advantage appears at 3-tool systematicity and in substitutability, not in 4-tool productivity, where ReAct's lenient call-everything behavior scores higher.

Interpretation: the in-sample "CGG" reading must control for chain depth; the paper's claims are limited to the matched-length comparisons highlighted above.

## 9 Seed-Level Variance (5 independent seeds)

Template agents are deterministic given the task seed; across seeds 42, 1, 2, 3, 4 the task instances differ, yielding genuine variance. Ranges are min–max over the 5 seeds; full per-seed values in `results/seed_variance.json`.

| Agent | Val | Systematicity | Productivity | Substitutability |
|---|---|---|---|---|
| ReAct | 0.248 [0.218, 0.268] | 0.429 [0.412, 0.445] | 0.513 [0.472, 0.542] | 0.251 [0.210, 0.295] |
| Native FC | 0.248 [0.218, 0.268] | 0.162 [0.145, 0.180] | 0.184 [0.175, 0.200] | 0.251 [0.210, 0.295] |
| Reflexion | 0.248 [0.218, 0.268] | 0.429 [0.412, 0.445] | 0.513 [0.472, 0.542] | 0.251 [0.210, 0.295] |
| Plan-Execute | 0.248 [0.218, 0.268] | 0.162 [0.145, 0.180] | 0.104 [0.097, 0.112] | 0.251 [0.210, 0.295] |
| **Chain-Aware** | **1.000** | 0.476 [0.438, 0.520] | 0.385 [0.293, 0.452] | **1.000** |

Table 9: TSR mean and min–max range over 5 independent task seeds (N per split as in Table **??**). ChainAware shows the largest productivity variance (0.293–0.452), so single-seed comparisons should use its 5-seed mean.

## 10 Hyperparameters and Configuration

- Random seed: 42 (default in all scripts)
- `max_steps`: 10 per task
- `n_synthetic_chains` (ChainAware): 50
- Bootstrap iterations: 1000

- Friedman / Wilcoxon $\alpha$: 0.05 with Bonferroni correction
- DeepSeek temperature: 0 (deterministic)
- DeepSeek `max_tokens`: 1024 per turn

## 11 Bootstrap 95% Confidence Intervals (TSR)

1000 resamples, seed=42. Full per-agent values in `results/bootstrap_ci.json`.

| Agent | Val | Sys. | Prod. | Sub. |
|---|---|---|---|---|
| ReAct | 0.268 [0.225, 0.312] | 0.428 [0.390, 0.467] | 0.542 [0.503, 0.580] | 0.250 [0.195, 0.310] |
| Native FC | 0.268 [0.225, 0.312] | 0.145 [0.117, 0.173] | 0.200 [0.168, 0.233] | 0.250 [0.195, 0.310] |
| Reflexion | 0.268 [0.225, 0.312] | 0.428 [0.390, 0.467] | 0.542 [0.503, 0.580] | 0.250 [0.195, 0.310] |
| Plan-Execute | 0.268 [0.225, 0.312] | 0.145 [0.117, 0.173] | 0.107 [0.083, 0.133] | 0.250 [0.195, 0.310] |
| ChainAware | 1.000 [1.000, 1.000] | 0.438 [0.400, 0.477] | 0.452 [0.412, 0.492] | 1.000 [1.000, 1.000] |
| DeepSeek ReAct | 0.130 [0.060, 0.200] | 0.060 [0.020, 0.110] | 0.040 [0.010, 0.080] | 0.140 [0.080, 0.210] |
| DeepSeek FC | 0.240 [0.160, 0.320] | 0.080 [0.030, 0.140] | 0.070 [0.030, 0.120] | 0.210 [0.130, 0.300] |

Table 10: Task Success Rate (TSR) with bootstrap 95% CIs per split.

Variance across independent training runs is not reported because all template agents are deterministic under the fixed seed 42; the bootstrap CIs above capture task-sampling variance. Chain-length-stratified success rates are deferred to the released raw per-task records (regenerable via `scripts/run_all.py -skip-llm`).

## 12 Reproducibility Checklist (NeurIPS-style)

- ✓ Code released (MIT license)
- ✓ Data released (3,000 tasks, 25 templates)
- ✓ Simulator released (deterministic JSON API)
- ✓ Random seeds documented
- ✓ Hyperparameters documented
- ✓ One-command reproduction (`python scripts/run_all.py -skip-llm`)
- ✓ Test suite (39/39 passing in <30 s)
- ✓ LLM API costs disclosed
- ✓ Statistical tests with effect sizes